

UVA 558

This problem involves determining whether a scientist can travel indefinitely back in time using a network of wormholes, which are modeled as directed edges between star systems with time shifts as weights. The core challenge is to detect if there exists a negative weight cycle reachable from the starting star system (node 0), as such a cycle would allow the scientist to loop through it repeatedly to go further back in time with each completion. The solution requires modeling the star systems as nodes in a directed graph where each wormhole from system x to system y with time change t represents a directed edge with weight t . Using the Bellman-Ford algorithm, we can identify negative cycles by relaxing all edges for n iterations (where n is the number of nodes); if any distance update occurs in the n th iteration, it confirms the presence of a negative cycle. The algorithm must ensure that the cycle is reachable from node 0, so we initialize distances with infinity except for the start node set to zero, and after $n-1$ iterations, an additional iteration that still reduces any distance reveals the desired cycle, leading to the output "possible"; otherwise, "not possible" is concluded.

Steps:

1. Understand the Problem Requirements

- We need to determine if there exists a cycle of wormholes where the total time change is negative
- This cycle must be reachable from our starting point (star system 0)
- A negative cycle would allow traveling back in time indefinitely

2. Model the Scenario as a Graph

- Represent each star system as a node in a directed graph
- Represent each wormhole as a directed edge with weight equal to the time change
- Note: Positive weights mean traveling forward in time, negative weights mean traveling backward in time

3. Apply Bellman-Ford Concept

- Start from star system 0 with initial time 0
- For each star system, keep track of the earliest time we can reach it
- Update these times by checking all possible wormhole travels

4. Detect Negative Cycles

- After finding the earliest arrival times to all reachable systems
- Check if we can further improve any arrival time by taking additional wormhole trips
- If any time can still be reduced, it indicates a negative cycle exists

5. Interpret the Result

- If a negative cycle is detected and it's reachable from start: output "possible"
- If no such cycle exists: output "not possible"

Input:

4 4

0 1 10

1 2 20

2 3 30

3 0 -60

Execution:

Edges stored as (u, v, weight):

- Edge 0: (0, 1, 10)
- Edge 1: (1, 2, 20)
- Edge 2: (2, 3, 30)
- Edge 3: (3, 0, -60)

Step 1: Initialization

- $\text{dist}[0] = 0$
- $\text{dist}[1] = \text{INF}$
- $\text{dist}[2] = \text{INF}$
- $\text{dist}[3] = \text{INF}$

Step 2: First Iteration (i=0) - Relax all edges

- Edge 0: $\text{dist}[0] + 10 = 10 < \text{INF} \rightarrow \text{dist}[1] = 10$
- Edge 1: $\text{dist}[1] + 20 = 10 + 20 = 30 < \text{INF} \rightarrow \text{dist}[2] = 30$
- Edge 2: $\text{dist}[2] + 30 = 30 + 30 = 60 < \text{INF} \rightarrow \text{dist}[3] = 60$
- Edge 3: $\text{dist}[3] + (-60) = 60 + (-60) = 0 \geq \text{dist}[0] \rightarrow \text{no change}$

After iteration 1: $\text{dist} = [0, 10, 30, 60]$

Step 3: Second Iteration (i=1) - Relax all edges

- Edge 0: $\text{dist}[0] + 10 = 10 \geq \text{dist}[1] \rightarrow \text{no change}$
- Edge 1: $\text{dist}[1] + 20 = 10 + 20 = 30 \geq \text{dist}[2] \rightarrow \text{no change}$
- Edge 2: $\text{dist}[2] + 30 = 30 + 30 = 60 \geq \text{dist}[3] \rightarrow \text{no change}$
- Edge 3: $\text{dist}[3] + (-60) = 60 + (-60) = 0 \geq \text{dist}[0] \rightarrow \text{no change}$

After iteration 2: $\text{dist} = [0, 10, 30, 60]$ (no changes)

Step 4: Third Iteration (i=2) - Relax all edges

- All edges: no changes (same calculations as iteration 2)

After iteration 3: $\text{dist} = [0, 10, 30, 60]$ (no changes)

Step 5: Fourth Iteration (i=3) - Check for negative cycles

- Edge 0: $0 + 10 = 10 \geq 10 \rightarrow \text{no change}$
- Edge 1: $10 + 20 = 30 \geq 30 \rightarrow \text{no change}$
- Edge 2: $30 + 30 = 60 \geq 60 \rightarrow \text{no change}$

- Edge 3: $60 + (-60) = 0 \geq 0 \rightarrow$ no change

Step 6: Analysis

- No distance updates occurred in the 4th iteration
- Therefore, no negative cycle reachable from node 0 exists

Output:

not possible

Pseudocode:

READ c

FOR each case:

 READ n, m

 edges = []

 FOR m times: READ u,v,w and store

 dist = [INF]*n

 dist[0] = 0

 REPEAT n times:

 FOR each edge (u,v,w):

 IF dist[u] + w < dist[v]:

 dist[v] = dist[u] + w

negative_cycle = false

FOR each edge (u,v,w):

 IF dist[u] + w < dist[v]:

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negative_cycle = true
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OUTPUT "possible" if negative_cycle else "not possible"
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